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Gender's Effect on Education and Money

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CS 458

Project 3

Location Information

Our study focuses on looking at how students experience the US education system. Each student has a unique set of circumstances that ultimately impact their experience with education. Location plays an important role when studying these experiences. Each student has plenty of factors that are affected by location when looking at the overall education experience.

When looking at the different experiences of students, it’s important to consider that the data is skewed by location. In this case, we are looking at the US only and different states have different populations. States with higher populations will have more representation in the data than states with lower populations. When looking at data from college onwards, the data is even more heavily skewed because of the locations of prominent schools.

With these things in mind, we looked at things like graduation rates of different states, testing scores, and overall performance of the students throughout their time in classes. Looking at the graduation rates among different states really helps highlight the skew in data caused by location. While the other analysis focused more on analyzing the actual experience of the students in different areas.

In the visualizations it’s very clear which states have more representation than others in the data. This is why we found using visualizations that highlight the differences. In the visualizations, it’s pretty clear that the differences are pretty drastic between the lower population states and the higher population states. We wanted to use a high color contrast to show how drastic the differences were in the visualizations. An example of this is shown in the choropleth map for average number of graduates for the different states. States like California that have a very high population with desirable colleges are a very dark red. Contrasting this, we have some of the midwest states. These states tend to have less desirable schools for most and have lower populations in general. These states are shown as a light yellow area.

Comparison of the Visualizations

In the field of data visualization, geospatial mapping techniques stand out for their ability to contextualize data within geographical parameters, thus providing a deeper insight into spatial relationships and patterns. This report delves into three distinct geospatial visualization techniques: Choropleth Maps, Heat Maps, and Hexagonal Binning. Each technique offers unique perspectives and solutions to specific visualization challenges, making them invaluable tools in data analysis and presentation.

Choropleth maps are a popular choice for visualizing data that is aggregated over predetermined areas, such as states or countries. These maps apply different colors or shades to areas based on the data values they represent, making it easy to identify patterns like population density or election results across different regions. The intuitiveness of choropleth maps comes from their direct correlation between color intensity and data magnitude, which allows viewers to grasp disparities and similarities across regions at a glance.

However, the effectiveness of choropleth maps can be compromised by the geographic size of the areas they depict. Larger areas might dominate the visual representation, potentially skewing the audience's perception of the data's importance. This issue is particularly pronounced in geographic regions where smaller areas might hold significant data values but are less visually prominent due to their size. Moreover, the choice of color scale can significantly affect readability and interpretation, requiring careful design to avoid misleading representations.

Transitioning from the region-specific choropleth, heat maps offer a fluid and dynamic visualization of data across geographic spaces without being bound by administrative boundaries. They use a gradient color scheme to represent data density or intensity, providing an effective means of identifying hot spots—areas of high concentration. This makes heat maps especially useful in environmental studies, disease mapping, or any field where understanding the concentration of phenomena is crucial.

Despite their strengths, heat maps rely heavily on interpolation and smoothing techniques to represent data points on a continuous surface. This can lead to inaccuracies if the underlying data is sparse or unevenly distributed, as the interpolation might create false impressions of homogeneity or exaggerate certain areas. Heat maps also require viewers to understand complex color gradients, which can be a barrier to interpretation without adequate legends and contextual explanation.

Hexagonal binning emerges as a powerful alternative to traditional map types, especially useful in visualizing large datasets. This technique overlays a map with a grid of hexagons, each representing data aggregated within its bounds. The hexagonal shape offers a balance between the geometric efficiency of circles (which cover the maximum area with the least perimeter) and the tessellation benefits of squares (which fit neatly together but do not approximate circles as closely).

Hexagonal binning minimizes the problems of arbitrary region boundaries and provides a uniform visual weight to all data points, making it highly effective for detailed, granular analysis. However, it comes with challenges in scalability and interactivity. Adjusting the size of hexagons to accommodate different zoom levels without losing data integrity requires sophisticated scaling algorithms, and the visual complexity of hexagonal maps can be daunting for users not familiar with this approach.

Overall, the choice between choropleth maps, heat maps, and hexagonal binning should be guided by the specific requirements of the data and the intended audience. Choropleth maps are best suited for straightforward regional comparisons, heat maps are ideal for visualizing data density and distribution, and hexagonal binning offers an advanced solution for managing large, granular datasets. By understanding the advantages and limitations of each geospatial visualization technique, data analysts and scientists can better tailor their visual presentations to meet analytical objectives and enhance audience understanding. Each technique has its place in the toolkit of a geospatial analyst, and the optimal choice often involves a combination of methods to fully elucidate complex spatial data relationships.

Chartio: A Comprehensive BI and Data Analytics Tool

Chartio is a highly effective business intelligence (BI) and data analytics tool, known for its robust data visualization capabilities. It integrates seamlessly with platforms like Amazon Redshift and Google BigQuery and supports CSV imports, making it versatile for various data sources. Chartio's standout feature is visual SQL, which allows users to graphically write and view SQL queries, simplifying complex query tasks. Users can then visualize their data using one of 15 chart options, enhancing data insights.

Pros:

1. Ease of Use: Visual SQL makes query writing intuitive, reducing the need for deep SQL knowledge.
2. Integration: Strong connectivity with major data sources like Amazon Redshift and Google BigQuery.
3. Visualization Options: Offers a variety of chart types for comprehensive data representation.

Cons:

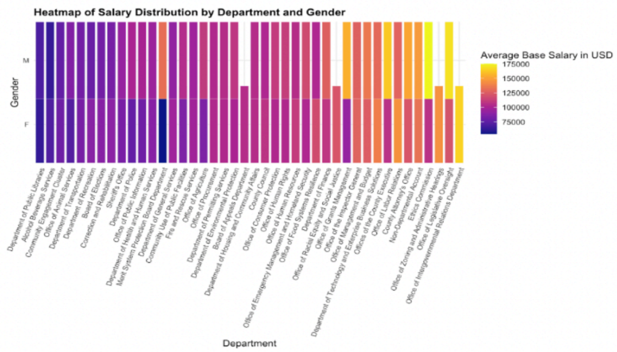
1. Learning Curve: While user-friendly, mastering all features may take time.
2. Cost: Depending on the plan, it can be expensive for small businesses.
3. Performance: Handling very large datasets can sometimes affect performance.

Suggestions for Improvement:

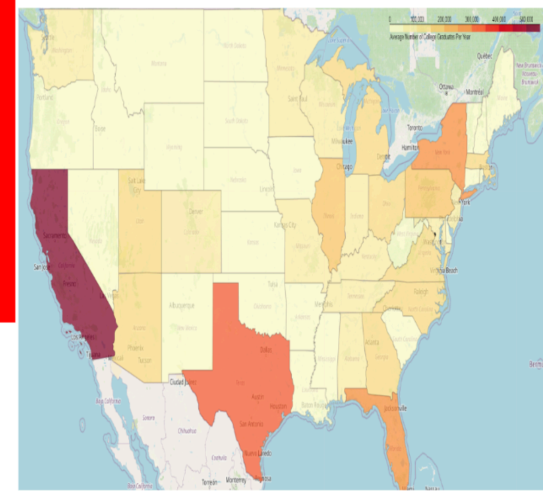
1. Enhanced Performance Optimization: Improving handling of large datasets to ensure consistent performance.
2. More Affordable Plans: Introducing tiered pricing to cater to smaller businesses and startups.
3. Advanced Customization: Allowing more customization options for visualizations to meet specific user needs.

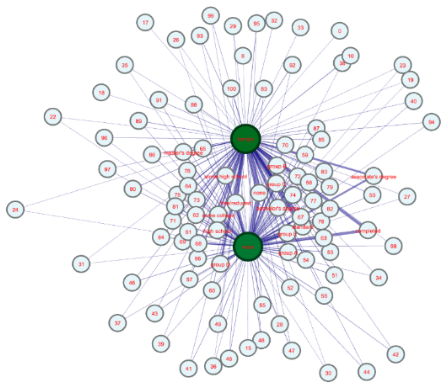
Overall, Chartio is a powerful tool that can significantly enhance data analysis and visualization efforts with its user-friendly features and strong integration capabilities.

Multiview Design



This heatmap on the left illustrates salary distributions by department and gender, using color intensity to represent average salaries ranging from $75,000 to $175,000. Each column corresponds to a department, differentiated by male (M) and female (F) rows. Notable disparities are evident in high-paying departments, where salary differences between genders become pronounced. This figure is crucial for our multi-view system, allowing an integrated analysis of gender-based salary dynamics within the organization.





The Network on the left visualization illustrates the relationships between various socio-economic factors and student performance. Nodes represent different variables, such as gender, parental education, lunch type, and performance scores,

This network on the right visualization represents socio-economic factors influencing student performance, filtered by edge weight between 76.924 and 334.0. Nodes represent variables such as gender, parental education, lunch type, and performance scores. Edges indicate relationships, with thicker edges denoting stronger correlations. Central nodes like "female" and "male" highlight their significant influence on various socio-economic factors and performance outcomes. The filtering emphasizes the most robust connections, providing a clearer view of the strongest relationships within the dataset.

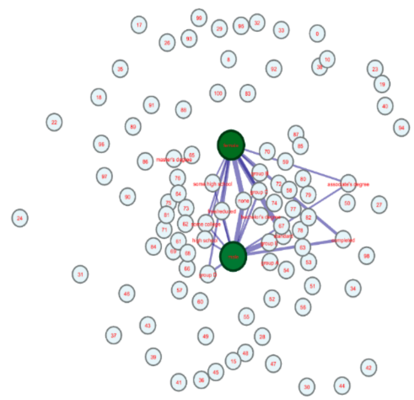


Figure 1: The inspiration behind our multi-view visualization design was to present our complex dataset, which is aimed at enhancing educational outcomes, in a straightforward and concise way. This was achieved by adhering to the principles outlined in the paper “Guidelines for Using Multiple Views in Information Visualization”. We implemented the rule of diversity, using various models such as heatmaps, map, and network diagrams to reinforce the data in the viewers’ memory. We emphasized on spacing, using perceptual techniques to direct the user’s attention to the appropriate diagrams in correlation with the right text boxes. By dividing it, we allow the user to process each image individually, leading to a better understanding of the dataset as a whole.